

BARSHINO, Kazakhstan

WMO: 353570

Lat: 49.683N

Lon: 69.517E

Elev: 1169

StdP: 14.09

Time Zone: 6.00 (E06)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.9	(c) -24.3	(d) -35.6	(e) 0.8	(f) -28.7	(g) -31.0	(h) 1.0	(i) -24.1	(j) 28.0	(k) 17.7	(l) 26.6	(m) 17.8	(n) 3.1	(o) 180	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 23.8	(c) 93.1	(d) 63.7	(e) 89.8	(f) 63.2	(g) 86.7	(h) 61.9	(i) 66.5	(j) 84.2	(k) 65.2	(l) 83.1	(m) 64.0	(n) 81.6	(o) 9.3	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.5	85.0	70.9	59.6	79.6	69.9	57.7	74.1	69.4	31.8	83.8	30.8	82.9	29.9	81.9	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.5	21.7	19.0	-31.2	98.3	6.2	2.7	-35.6	100.2	-39.2	101.8	-42.7	103.4	-47.2	105.3		
			-31.4	70.4	6.1	2.9	-35.8	72.5	-39.4	74.2	-42.8	75.8	-47.3	78.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.7	0.8	1.5	21.0	45.7	57.0	67.8	70.2	68.3	55.8	40.8	23.8	8.8	
	DBStd	27.32	13.90	13.74	12.39	9.45	7.76	6.88	6.33	6.81	8.10	8.97	12.92	14.98	
	HDD50	6391	1524	1358	900	184	25	0	0	0	32	303	787	1278	
	HDD65	10115	1989	1778	1363	578	264	51	23	45	295	750	1236	1743	
	CDD50	2250	0	0	2	56	242	534	625	567	206	18	0	0	
	CDD65	498	0	0	0	0	16	134	183	146	19	0	0	0	
	CDH74	7102	0	0	1	44	408	1904	2369	2010	360	6	0	0	
	CDH80	2828	0	0	0	5	80	772	1015	834	122	0	0	0	
Wind	WSAvg	8.0	7.6	7.6	9.3	9.1	8.9	7.8	8.0	7.2	6.9	7.5	8.1	8.2	
Precipitation	PrecAvg	9.2	0.6	0.6	0.5	0.7	1.0	1.5	1.2	0.7	0.5	0.7	0.7	0.6	
	PrecMax	12.9	1.1	1.2	1.3	1.6	2.2	4.3	3.0	3.0	1.9	2.2	2.4	1.9	
	PrecMin	4.1	0.2	0.2	0.1	0.0	0.1	0.1	0.0	0.0	0.1	0.1	0.1	0.0	
	PrecStd	2.5	0.2	0.2	0.4	0.5	0.6	1.2	0.8	0.7	0.5	0.6	0.5	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.6	34.0	56.7	79.5	86.3	94.4	96.8	97.2	90.5	73.1	53.8	35.4	
		MCWB	31.4	31.6	47.8	56.9	58.5	64.0	66.2	65.0	61.9	54.3	49.4	32.4	
	2%	DB	30.6	30.2	46.1	73.3	82.4	91.1	93.2	92.2	82.8	66.8	46.7	33.3	
		MCWB	28.6	27.8	41.6	54.2	58.2	64.4	63.9	63.6	59.0	49.7	41.3	30.8	
	5%	DB	26.2	27.4	40.1	67.4	78.7	88.1	89.7	88.0	77.3	61.2	42.2	31.1	
		MCWB	24.5	25.5	36.2	51.0	57.6	62.4	63.4	62.4	56.8	47.4	38.3	28.9	
	10%	DB	22.1	23.5	35.9	62.5	75.2	84.7	86.1	84.6	72.3	55.4	38.5	28.5	
		MCWB	20.5	21.9	32.8	49.3	55.9	61.4	62.6	61.1	54.1	45.0	35.1	26.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.7	31.5	48.1	58.1	63.4	70.1	68.5	67.0	64.7	54.6	50.5	32.9
			MCDB	33.5	33.4	53.2	75.8	76.6	86.7	86.0	85.5	85.4	69.3	52.6	34.5
2%		WB	28.6	28.2	41.7	55.4	60.8	66.2	66.7	65.6	59.9	51.0	42.3	31.1	
		MCDB	30.5	30.3	46.0	70.0	76.2	83.6	83.9	84.9	80.2	62.7	45.8	33.1	
5%		WB	24.7	25.6	36.0	52.6	59.0	64.7	65.4	64.2	57.0	48.4	38.4	29.0	
		MCDB	26.2	27.5	39.7	64.7	74.2	82.5	82.7	83.1	73.6	58.8	41.6	31.1	
10%		WB	20.6	21.9	33.1	50.1	57.2	63.1	64.2	62.6	54.8	46.0	35.3	26.4	
		MCDB	22.1	23.6	35.5	60.2	73.3	79.9	81.2	81.3	70.5	54.6	38.5	28.5	
Mean Daily Temperature Range	5% DB	MDBR	16.5	18.6	16.6	21.8	26.1	25.0	23.8	25.0	24.9	20.3	14.5	15.6	
		MCDBR	12.6	14.6	17.5	31.7	31.6	30.6	29.8	31.5	32.9	31.3	17.0	13.9	
		MCWBR	12.4	13.8	12.7	15.5	13.7	11.5	10.1	12.1	14.9	17.3	12.0	12.7	
	5% WB	MCDBR	12.9	14.6	16.1	27.6	27.7	25.6	25.1	27.3	28.7	27.8	15.1	12.9	
		MCWBR	12.8	13.8	12.3	15.1	13.4	11.2	9.7	11.8	14.0	16.8	11.7	12.0	
Clear-Sky Solar Irradiance	taub	0.296	0.331	0.382	0.413	0.415	0.412	0.429	0.401	0.353	0.342	0.328	0.299		
	taud	2.078	2.001	2.024	2.120	2.182	2.254	2.218	2.281	2.364	2.313	2.189	2.104		
	Ebn at Noon	226	246	255	262	268	269	262	264	267	246	213	205		
	Edh at Noon	27	38	44	45	44	42	43	38	31	28	24	23		
All-Sky Solar Radiation	RadAvg	497	868	1269	1599	2010	2119	1977	1787	1360	773	385	338		
	RadStd	50	52	134	145	137	138	107	132	88	101	45	50		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page